EXPERIMENT 8

Academic Year: 2026-27 **Year and Sem: BE VIII**
Subject: Recommendation Systems Lab **Course Code: CSDOL8022**

Aim: Implementation of recommendation system using Ensembled approach

Theory: An ensemble recommendation system combines multiple individual recommendation models—such as collaborative filtering, content-based filtering, and matrix factorization—to generate superior, more accurate, and diverse recommendations. It enhances predictive performance (reducing MSE/RMSE) by leveraging the strengths of diverse algorithms, effectively solving complex recommendation tasks and mitigating overfitting.

Ensemble Recommendation Algorithm

Input:

- Ratings dataset
- Movies dataset
- Weights (w_1, w_2, w_3)

Steps:

1. **Construct User-Item Matrix**
   - Pivot ratings table
   - Fill missing values with 0
2. **Compute Content Similarity**
   - Apply TF-IDF on genres
   - Compute cosine similarity
3. **Train Collaborative Model**
   - Apply Truncated SVD
   - Learn latent features
4. **Compute CF Scores**
   - Find similarity between users
   - Predict movie scores
5. **Compute CB Scores**
   - Average similarity for each movie
6. **Compute Popularity Scores**
   - Calculate mean rating per movie
7. **Combine Scores**

   Score=w1(CF)+w2(CB)+w3(Popularity)
8. **Sort and Recommend Top-N Movies**

Advantages

Improves recommendation accuracy
Handles cold start problem
Combines strengths of multiple models
More robust than single model

Limitations

Computationally expensive
Requires tuning of weights
Complex implementation

Conclusion: The ensemble approach improves recommendation quality by integrating collaborative filtering, content-based filtering, and popularity-based methods. It overcomes the limitations of individual techniques and provides more reliable and accurate recommendations. Proper selection of weights plays a crucial role in optimizing system performance.